Assignment 1
Optimal Modelling, Control, and Estimation of a Cart-Pendulum System

Deadline: Friday 27th March 2026.

Q1. Least-Square Based System Model Identification of a Cart-Pendulum System (50 marks)

By using the Euler-Lagrange method, and considering $q_1 = x_c$ and $q_2 = \alpha$ as generalised coordinates, the cart-pendulum system in Fig. 1 when pointing downwards can be described by:

$$K_f \cdot v_m = (M + m)\,\ddot{x}_c + b_c\dot{x}_c + m\ell\ddot{\alpha}\cos(\alpha) - m\ell\dot{\alpha}^2\sin(\alpha) \tag{1a}$$

$$0 = m\ell^2\ddot{\alpha} + b_p\dot{\alpha} + m\ell\ddot{x}_c\cos(\alpha) + gm\ell\sin(\alpha). \tag{1b}$$

where v_m stands for the dc-motor armature voltage, x_c and v_C are the cart position and speed, α and ω are the pendulum angle and angular speed, M and m are the cart and pendulum mass, and b_c b_p the cart and b_p the pendulum friction respectively. For this assignment, we assume no knowledge of the system parameters. Moreover, the only accessible information are the applied voltage to the dc-motor, v_m, the measured cart position, x_c, and pendulum angle, α.

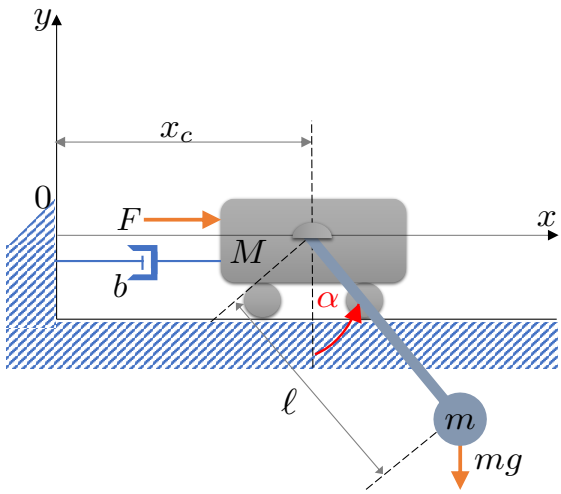

Figure 1: Cart-pendulum system pointing downward.

a) By assuming a small angle approximation, $\alpha \approx 0$, and a slow angular speed, $\dot{\alpha}^2 \approx 0$; and considering $\boldsymbol{x}(t) = \begin{bmatrix} x_c(t) & \alpha(t) & v_c(t) & \omega(t) \end{bmatrix}^\intercal$, $\boldsymbol{y}(t) = \begin{bmatrix} x_c(t) & \alpha(t) \end{bmatrix}^\intercal$, and $u(t) = v_m(t)$; obtain an analytic continuous-time state-space model of the cart-pendulum system, i.e.:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}_c\boldsymbol{x}(t) + \boldsymbol{B}_c u(t) \tag{2a}$$

$$\boldsymbol{y}(t) = \boldsymbol{C}_c\boldsymbol{x}(t) \tag{2b}$$

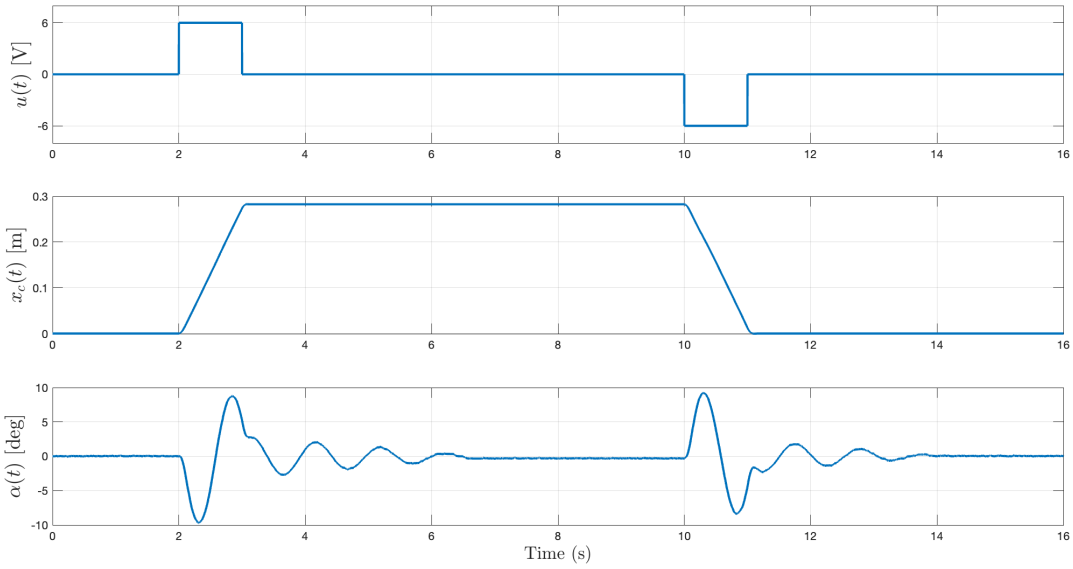

Figure 2: Experimental data from a pulse response of a cart-pendulum system pointing downwards.

Provide analytic expressions for matrices $\boldsymbol{A}_c$, $\boldsymbol{B}_c$, and $\boldsymbol{C}_c$.

b) By applying a forward-Euler discretisation method to the system (2), $\dot{\boldsymbol{x}} \approx \frac{\boldsymbol{x}(k+1)-\boldsymbol{x}(k)}{t_s}$, obtain an analytic discrete-time state space model, i.e.:

$$\boldsymbol{x}(k+1) = \boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{B}u(k) \tag{3a}$$
$$\boldsymbol{y}(k) = \boldsymbol{C}\boldsymbol{x}(k) \tag{3b}$$

Provide analytic expressions for matrices $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$.

c) By inspecting the system model (3), determine the number of unknown model parameter that need to be identified, $\boldsymbol{\theta} = \begin{bmatrix}\theta_1 & \cdots & \theta_r\end{bmatrix}^\mathsf{T}$. Then, rearrange (3) to obtain the following system identification relationship:

$$\boldsymbol{b}_\ell = \mathcal{A}_\ell \boldsymbol{\theta} \tag{4}$$

Provide analytic expressions for $\boldsymbol{\mathcal{A}}_\ell$, and $\boldsymbol{b}_\ell$ in terms of $\boldsymbol{x}(k)$ and $u(k)$.

d) One of the benefits using a cart-pendulum pointing downwards is that the equilibrium point, $\alpha = 0$, is stable. This allows one to excite the system by applying a pulse input and then record its response in terms of position displacement and angle oscillation. In your lab setup, apply a positive and negative pulse to displace the cart and introduce an oscillation to the pendulum. You should obtain an experimental result similar to the one depicted in Fig. 2. This will result in the following data-set:

$$\boldsymbol{U} = \begin{bmatrix}u(k) & u(k+1) & \cdots & u(k+N)\end{bmatrix} \tag{5a}$$
$$\boldsymbol{Y} = \begin{bmatrix}\boldsymbol{y}(k) & \boldsymbol{y}(k+1) & \cdots & \boldsymbol{y}(k+N)\end{bmatrix} \tag{5b}$$

e) Since the only available data is the system output $\boldsymbol{y}(k)$ and its input $u(k)$, then it is necessary to estimate the full system state, $\hat{\boldsymbol{x}}(k)$. To achieve that, design an implement a steady-state Kalman smoother:

$$\hat{\boldsymbol{x}}(k+1) = \boldsymbol{A}_f\hat{\boldsymbol{x}}(k) + \boldsymbol{K}_f\left(\boldsymbol{y}(k) - \hat{\boldsymbol{y}}(k)\right) \tag{6a}$$
$$\hat{\boldsymbol{y}}(k) = \boldsymbol{C}\hat{\boldsymbol{x}}(k) \tag{6b}$$

Prove that system (6) is observable. Then, explain how you have tuned the covariance matrices $\boldsymbol{Q}_f$ and $\boldsymbol{R}_f$ to obtain $\boldsymbol{K}_f$. The implementation of this Kalman smoother will lead to the following smooth data-set:

$$\boldsymbol{U} = \begin{bmatrix} u(k) & u(k+1) & \cdots & u(k+N) \end{bmatrix}, \tag{7a}$$
$$\hat{\boldsymbol{X}} = \begin{bmatrix} \hat{\boldsymbol{x}}(k) & \hat{\boldsymbol{x}}(k+1) & \cdots & \hat{\boldsymbol{x}}(k+N) \end{bmatrix}. \tag{7b}$$

f) Implement in MATLAB an offline Least-Square algorithm by expanding (4) to account for the whole smooth data-set in (7), i.e.:

$$\boldsymbol{\mathcal{A}} = \begin{bmatrix} \mathcal{A}_1 \\ \vdots \\ \mathcal{A}_\ell \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_\ell \end{bmatrix}.$$

Provide your identified parameters $\hat{\boldsymbol{\theta}}$.

g) Implement in MATLAB an offline Recursive Least Square algorithm to obtain $\hat{\boldsymbol{\theta}}(k)$.

$$\boldsymbol{K}_i(k) = \boldsymbol{P}_i(k-1)\boldsymbol{\mathcal{A}}^\intercal(k)\left(\boldsymbol{\mathcal{A}}(k)\boldsymbol{P}_i(k-1)\boldsymbol{\mathcal{A}}^\intercal(k) + \boldsymbol{R}_i\right) \tag{8a}$$
$$\hat{\boldsymbol{\theta}}(k) = \hat{\boldsymbol{\theta}}(k-1) + \boldsymbol{K}_i(k)\left(\boldsymbol{b}(k) - \boldsymbol{\mathcal{A}}(k)\hat{\boldsymbol{\theta}}(k)\right) \tag{8b}$$
$$\boldsymbol{P}_i(k) = \left(\boldsymbol{I} - \boldsymbol{K}_i(k)\boldsymbol{\mathcal{A}}(k)\right)\boldsymbol{P}_i(k-1) \tag{8c}$$

Compare these identified parameters with the ones obtain in f).

Challenge:

h) Implement in your control-board the same Recursive Least Square algorithm to identify the system parameters $\hat{\boldsymbol{\theta}}(k)$ in real-time.

Q2. Linear Quadratic Regulator Design for a Cart-Pendulum System (40 marks)

a) For the identified discrete-time model in (3) considering the identified parameters $\hat{\boldsymbol{\theta}}$. Prove that the system is controllable. Then, design an LQR strategy:

$$u(k) = -\boldsymbol{K}\left(\boldsymbol{x}(k) - \boldsymbol{x}^\star\right) + u^\star. \tag{9}$$

Test/simulate your design in MATLAB/Simulink by considering that all system estates, $\boldsymbol{x}(k)$, are measured.

Remark 1. *initially, when designing and tuning an LQR, it is recommended to implement a simulation assuming all system states can be measured, i.e.* $\boldsymbol{y}(k) = \begin{bmatrix} x_c(k) & \alpha(k) & v_c(k) & \omega(k) \end{bmatrix}^\intercal$. *When you obtain a desired response, you can later design an observer to estimate all system states.*

Remark 2. *It is important you test different values for the weighting matrices* $\boldsymbol{Q}$ *and* $\boldsymbol{R}$ *to obtain a good trade-off between closed-loop dynamic response and control effort.*

b) Prove that the system (3) evaluated with the identified parameters $\hat{\boldsymbol{\theta}}$ is observable. Then, design a Kalman filter to estimate in real-time the system estates:

$$\hat{\boldsymbol{x}}(k+1) = \boldsymbol{A}\hat{\boldsymbol{x}}(k) + \boldsymbol{B}u(k) + \boldsymbol{K}_f\left(\boldsymbol{y}(k) - \hat{\boldsymbol{y}}(k)\right) \tag{10a}$$

$$\hat{\boldsymbol{y}}(k) = \boldsymbol{C}\hat{\boldsymbol{x}}(k) \tag{10b}$$

Then, test/simulate your designed optimal control strategy in MATLAB/Simulink by considering by combining both LQR and Kalman filter, i.e., only the measurements for cart position, x_c, and pendulum angle, α, are available: $\boldsymbol{y}(k) = \begin{bmatrix} x_c(k) & \alpha(k) \end{bmatrix}^\intercal$,

Remark 3. *Note that this filter/observer uses the identified model in* (3)*. Therefore, you cannot assume that the same Kalman gain used in* (6) *will work for this filter. Therefore, you have to tune the system covariance* $\boldsymbol{Q}_f$ *to design* $\boldsymbol{K}_f$ *for this observer.*

Remark 4. *It is important you test different values for* $\boldsymbol{Q}_f$ *to find the one give you a good trade-off between dynamic response and filtering capability.*

Challenges:

c) Implement in your control-board your designed optimal control strategy (LQR plus Kalman filter) for the cart-pendulum system in the lab.

d) Implement in your control-board both your Recursive Least Square algorithm and your designed optimal control strategy (LQR plus Kalman filter) for the cart-pendulum system in the lab.

Conclusions (10 marks):

Provide an overall conclusion about the whole process to optimally identify a system model and to design an optimal control strategy for the cart pendulum system.

References:

[1] R. P. Aguilera, *"Lecture Notes on Nominal Feedback Control"*, University of Technology Sydney, Australia, 2026.

[2] R. P. Aguilera, *"Slides on Optimal Filtering and System Identification"*, University of Technology Sydney, Australia, 2026.

[3] R. P. Aguilera, *"Slides on Optimal Control"*, University of Technology Sydney, Australia, 2026.